On the extreme scaling of transistors with monolayer MoS_2 channel

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Abstract

2D transition metal dichalcogenides (TMDs) show promise for transistor scaling, but their on-scale performance had not been proven yet. This work demonstrates contact length (L_C) scaling while holding a low contact resistance (R_C) down to 11 nm. Channel length (L_{CH}) scaling shows I_{ON} can increase down to at least 12 nm with low R_C. The very scaled (L_{CH} = 19 nm and L_C = 12 nm) MoS_2 transistor with Sb-based metal contact has current density of ~1130 μA/μm at V_{DS} = 1 V, and a low R_C of ~190 Ω·μm. These scaled transistors, processed within a back-end-of-line (BEOL) thermal budget, do not exhibit subthreshold swing (S.S.) degradation or observable drain-induced barrier lowering (DIBL) down to L_{CH} = 12 nm.

Introduction

Over several decades, the continuous miniaturization of transistors has propelled the advancements in Si logic technology. Low-dimensional materials, particularly two-dimensional (2D) transition metal dichalcogenides (TMDs), are interesting candidates to enable scaling to the smallest size [1]-[4]. Therefore, understanding the scaling limits of transistors with monolayer TMD channels is critical. Modeling analysis projects that ON-current (I_{ON}) at V_{DD} = 1 V increases as the channel length (L_{CH}) scales down to 10 nm (see Fig. 1). However, I_{ON} is expected to decrease when scaling contact length (L_C) as contact resistance is projected to increase. Unsurprisingly, double-gated (DG) devices can offer higher I_{ON} and better electrostatic control than back-gated (BG) devices of the same footprint and potentially enable device scaling to less than 35 nm. Until now, it has not been proven in experiments that both L_{CH} and L_C can be scaled in 2D transistors with good performance. This work reports the first extremely scaled transistors of monolayer MoS_2 channel in which current density scaled upon channel length reduction at a relevant L_C = 12 nm.

Results and Discussion

Back-gated devices fabricated through a simple method enable L_C scaling experimentally from 20 nm to 0 nm (edge-only contact). Representative TEM images verify the formation of such scaled L_C in Fig. 2. Note that only those contact regions with 2D underneath are effective for current injection and thus defined as L_C. With this method, physical metal width is ~ 300nm and >> the transfer length. As a result, metal line resistance does not impede the device parameter extraction. Top contact is dominating the current injection with more than two orders of magnitude higher current comparing to edge-only contact in Fig. 3. Therefore, present contact geometry (top + edge) represents well the conventional top contact. It is worth mentioning that the edge portion is not yet optimized, and it could potentially be used to further lower R_C. To quantify the tradeoff of R_C increasing with L_C scaling, the transfer characteristics of L_C from 20 nm to 2.4 nm in Fig. 4 are analyzed. Nearly unchanged I_{ON} is observed down to 11 nm. Transmission line method (TLM) is employed to separate R_C and mobility at given L_C and compared with TCAD modeling extracted values in Fig. 5 and 6. For details of the model, see ref. 5. R_C remains ~ 300 Ω·μm down to 11 nm and thus L_C of 12 nm will be used for the following performance assessment through L_{CH} scaling.

The electrical characteristics of a device with device length (DL) of 31 nm (L_C 12 nm and L_{CH} 19 nm) are shown in Fig. 7. Solid lines represent the TCAD model, developed in our previous work [5]. The model describes well the transfer characteristics from subthreshold region to on-state and output characteristics from low to high field. R_C of 186 Ω·μm and injection velocity at the top of the barrier of 4.25×10^6 cm/s can, therefore, be extracted. TEM images and corresponding EDS mapping identify the L_C and L_{CH}, where the L_C is determined by the overlapping of 2D and metal contact, and as a result, DL of 33 nm is verified. It also confirms no residue at the interface of antimony and 2D revealing direct contact.

Current density shows a scaling trend upon L_{CH} scaling down to at least 12 nm as demonstrated experimentally in Fig. 9a. This gives confidence in the channel scaling potential with ultra-thin body 2D channel material. Electrostatic control is also found to be unaltered with S.S. and DIBL showing no observable increase with decreasing L_{CH} in Fig. 9b,c. The overall S.S. of the devices is non-ideal because of the non-optimized interface between channel and bottom dielectric layer. Extracted D_{it} is larger than 3×10^{12} cm^{-2}. We ascribe the observed variability to local D_{it} changes which could possibly be resolved through surface treatment or channel quality improvement. Note that extracted DIBL values could be impacted by traps and associated hysteresis from the substrate: DIBL×ΔV_{DS} is smaller or comparable to the hysteresis size.

In Fig. 10a, we benchmark the I_{ON} results with literature. Current increasing with DL scaling has not been observed before due to high R_C. This work is the first to demonstrate devices with scaled sub-20 nm L_{CH} and 12nm L_C with the nearly Ohmic contact, achieving high I_{ON} above 900 μA/μm at V_{DS} = 1 V. The performance gap between experiments and TCAD projected I_{ON} stems from non-optimized gate stack modules and 2D channel quality. In Fig. 10b, the measured (filled yellow squares) and model-projected (open green squares) I_{ON} is plotted against S.S. at V_{DD} of 1 V with I_{OFF} = 20 nA/μm. The most critical step needed is gate stack improvement through D_{it} reduction to 5×10^{11} cm^{-1} and EOT scaling down to 0.7 nm.

Summary

We report the first on-scale transistors with monolayer MoS_2 channel at L_{CH} = 19 nm and L_C = 12 nm with I_{ON} reaching up to ~1100 μA/μm at V_{DS} = 1 V with extracted R_C ~190 Ω·μm. The devices show I_{ON} increasing upon L_{CH} scaling from ~33 nm to ~12 nm without observable degradation in S.S. and DIBL, albeit at relatively high starting S.S. Comprehensive verification of L_C scaling from 20 nm to 2.4 nm is provided for the first time as well for nearly ohmic contacts. The Sb-based metal contact can enable very scaled contacts with a relatively low R_C ~300 Ω·μm at L_C ~ 11 nm. These findings validate the scaling potential of monolayer TMD materials to sub-40 nm device length. While the results are very encouraging for device length scaling, reduction of gate stack defects as well as further EOT scaling are required, crucial elements for performance.

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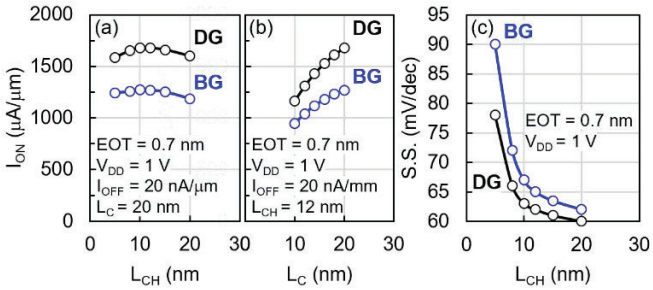

Fig. 1. Model of I_{ON} as a function of (a) channel length (L_{CH}) and (b) contact length (L_C). (c) S.S. versus L_{CH}. V_{DD} = 1 V, EOT = 0.7 nm, I_{OFF} = 20 nA/μm. BG is back gate overlapping with contact. DG is double gate gating channel from both sides. I_{ON} is normalized to the same 2D channel width.

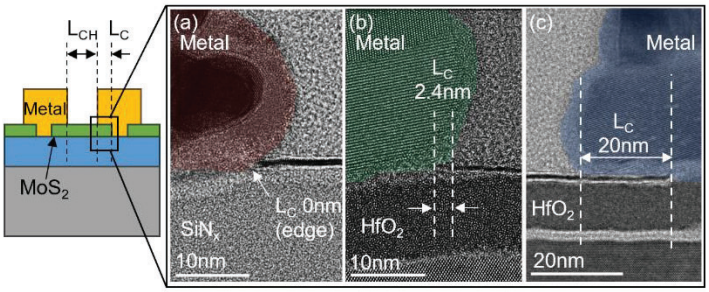

Fig. 2. (a) Device sketch and representative TEM images of L_C = (a) 0 i.e. edge-only contact), (b) 2.4, and (c) 20 nm. Note that only the regions where both metal and the 2D material overlap form effectively a contact and are thus defined as L_C.

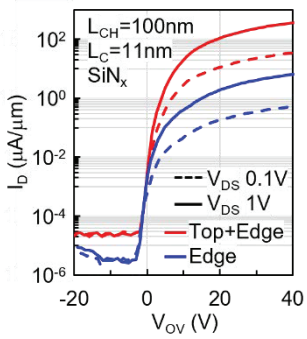

Fig. 3. Top (L_C =11 nm) + edge contact and edge-only contact to MoS_2 on 100 nm SiN_x. Edge contributes little to overall transport.

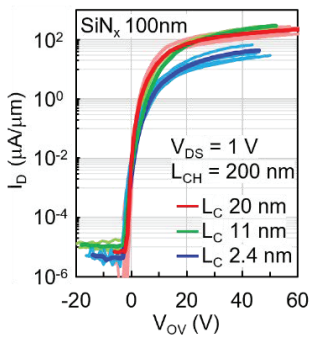

Fig. 4. Similar I_{ON} is observed at L_C = 11 and 20 nm while L_C = 2.4 nm yields lower current density. Thick/light lines are the average/ raw data.

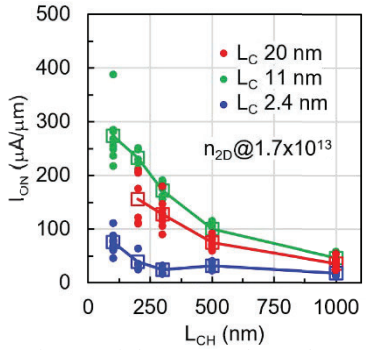

Fig. 5. I_{ON} is lower at L_C shorter than 11 nm. The non-monotonic dependence of I_{ON} on L_C is ascribed to different μ as different L_C devices are fabricated on different samples.

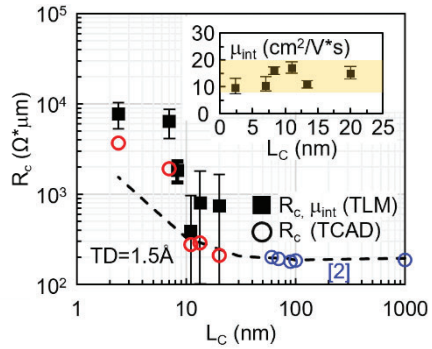

Fig. 6. R_C as a function of L_C. Filled/open symbols are R_C extracted from TLM and TCAD model, respectively. Dashed line is TCAD model R_C with TD = 1.5 Å, SBH = 0 eV, and μ= 20 cm^2/Vs.

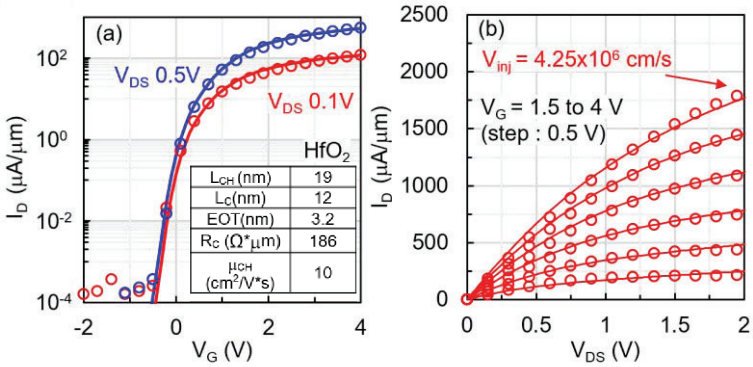

Fig. 7. (a) Transfer and (b) output characteristics of a device with EOT 3.2 nm. Symbols are experimental data, lines are TCAD model. R_C ~190 Ω·μm, μ =10 cm^2/Vs, and V_{inj} = 4.25×10^6 cm/s.

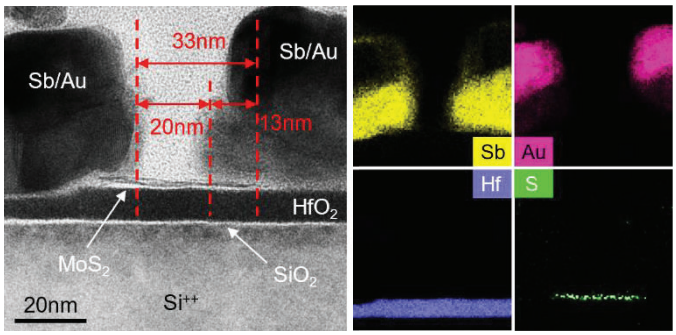

Fig. 8. Representative TEM images and corresponding EDX mapping show L_{CH} and L_C of 20 and 13 nm on the EOT 3.2 nm substrate. Device length (DL) defined as the sum of L_{CH} and L_C is 33 nm.

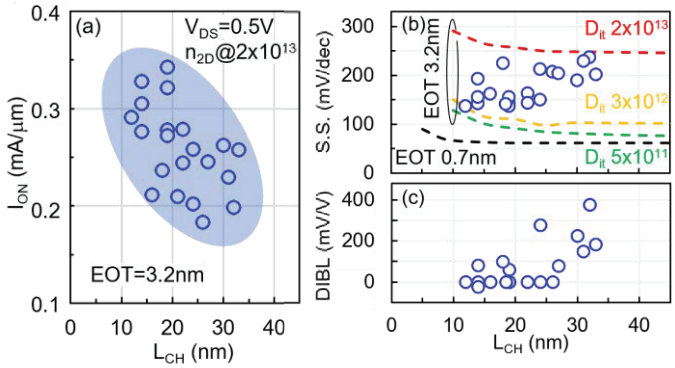

Fig. 9. (a) On-current at n_{2D} of 2×10^{13} cm^{-2} versus L_{CH} at V_{DS} 0.5 V. (b) S.S. and (c) DIBL extracted at I_D = 10 nA/μm. TCAD modeled S.S. at several D_{it} levels are shown in dashed lines. Note that DIBL×ΔV_{DS} is either smaller than or comparable to the corresponding hysteresis.

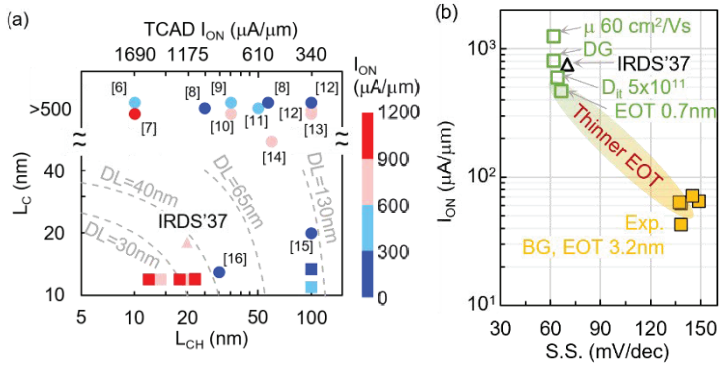

Fig. 10. (a) I_{ON} benchmark at V_{DS} = 1 V vs L_{CH} and L_C. Squares are present work, circles are literature. Top axis is the corresponding TCAD projection of I_{ON} at L_C = 12. Reference7 has I_{ON} = 1120 μA/μm at V_{DS} = 0.7 V with tri-layer MoS_2. (b) Path to improved I_{ON}: projections are based on experiment at V_{DD} = 1 V, and I_{OFF} = 20nA/μm.

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